Amrapali Opencast Mine

🏭 Introduction

The **Amrapali Opencast Mine** is a major coal mining project operated by Central Coalfields Limited, a subsidiary of Coal India Limited. It is one of the most important mines in the North Karanpura coalfield and plays a key role in supplying coal to power plants in eastern and northern India.

Commissioned in the early 2010s, Amrapali OCP has rapidly developed into a high-capacity mine with an annual production capacity of around **15–20 million tonnes**. The mine is highly mechanized, using shovel–dumper combinations, drills, and modern surface mining equipment. It is often studied for its efficient operations and contribution to India's growing energy demand.

🌍 Geographical Overview – Amrapali Opencast Mine

The Amrapali mine is located in the Chatra district of Jharkhand, within the **North Karanpura coalfield**, one of the largest coal-bearing regions in India.

📍 Location & Setting

- Situated in **Tandwa block** of Chatra district
- Part of the **North Karanpura basin**
- Well connected by road; rail connectivity is being expanded for coal evacuation

🌄 Physical & Geological Features

- Lies in a plateau region with undulating terrain
- Contains **thick coal seams of Gondwana formation**
- Coal type: **non-coking thermal coal**, suitable for power generation

🌊 Nearby Natural Features

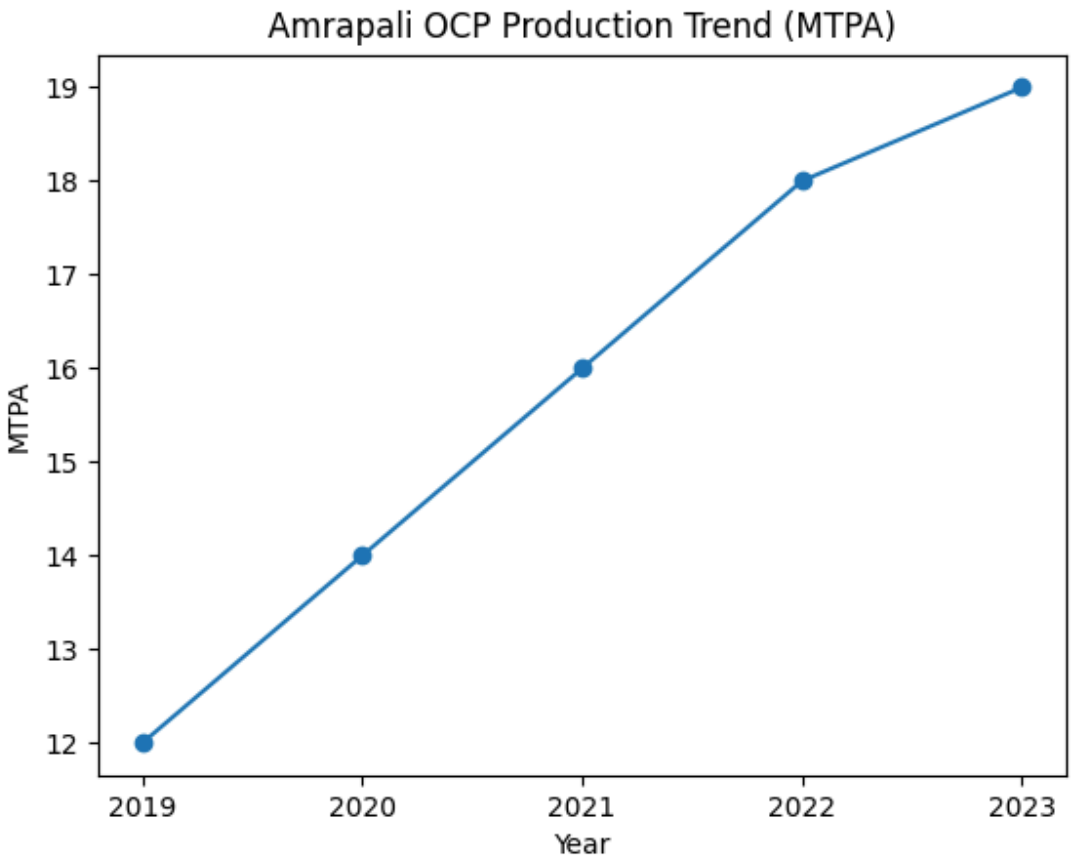
- Close to tributaries of the Damodar River basin
- Forested region with significant biodiversity (affected by mining activities)

🌦️ Climate

- Tropical climate
- Hot summers (up to ~45°C)
- Moderate to heavy monsoon rainfall

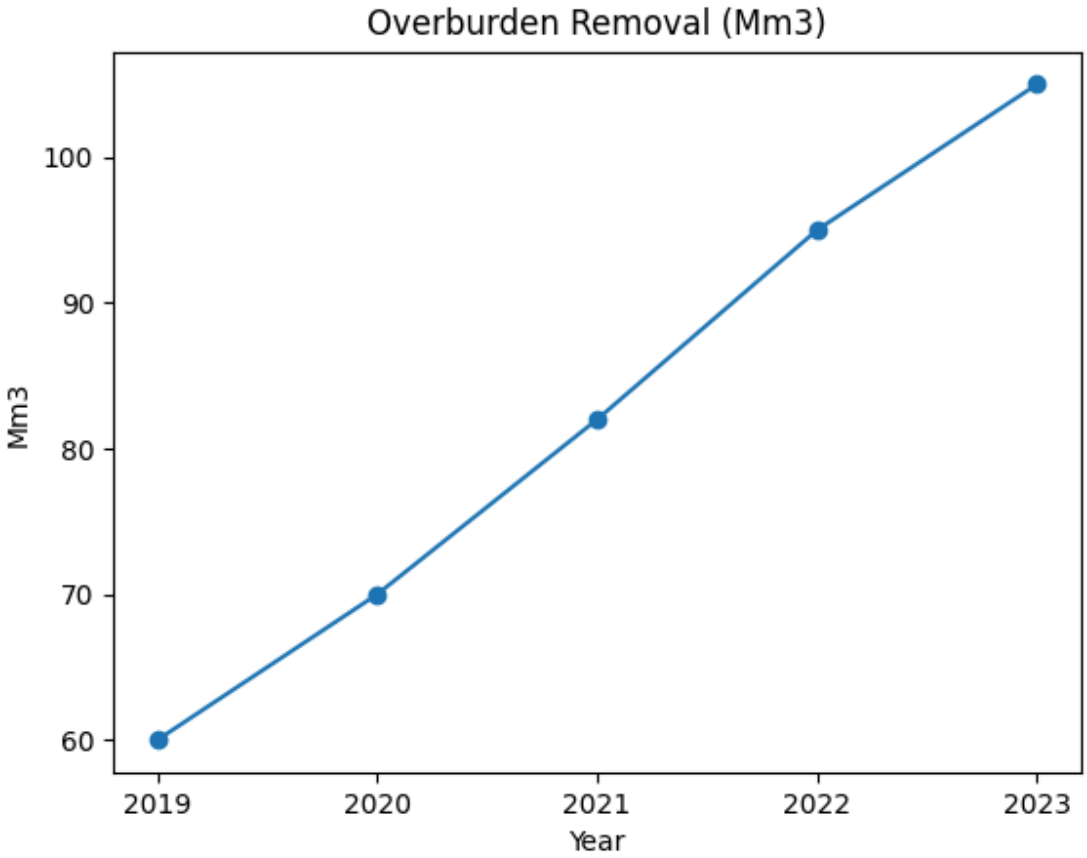
- Cool winters

📊 1. Production Trend (MTPA)



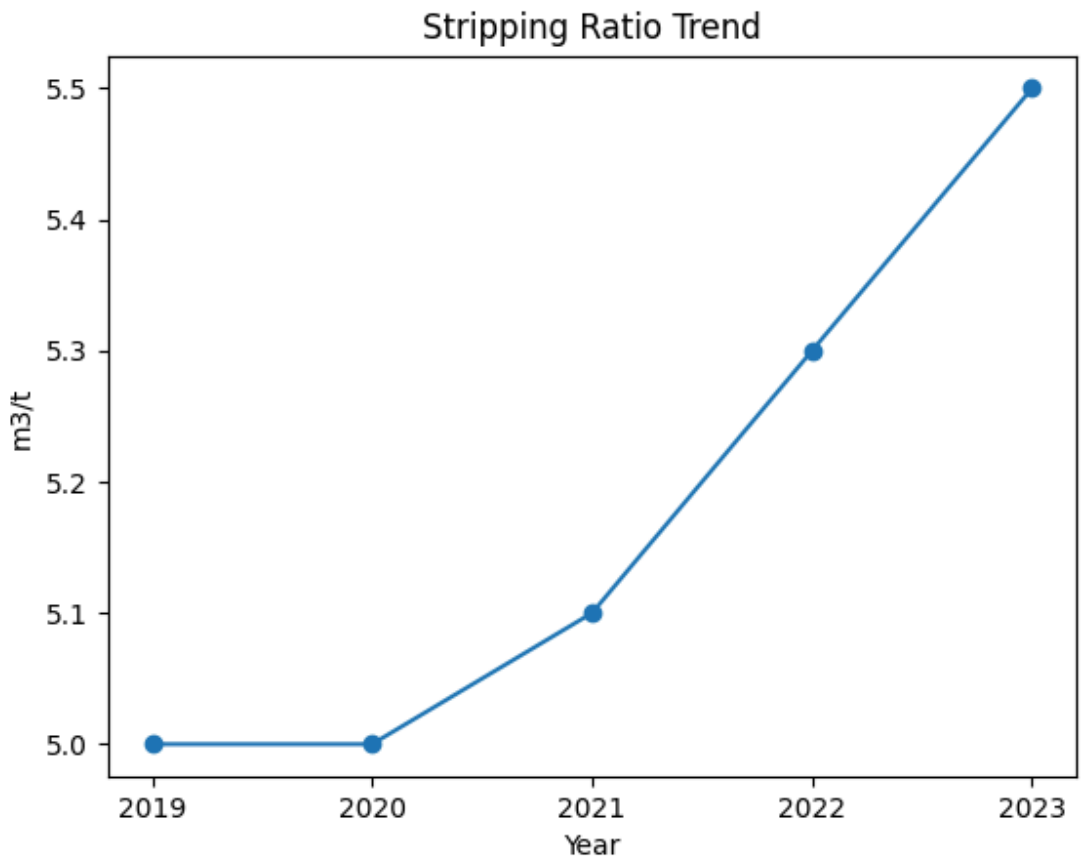
This chart shows the steady increase in coal production at Amrapali OCP, operated by Central Coalfields Limited.
It reflects expansion of mining operations and improved efficiency over time.
Significance: Useful for production forecasting, mine planning, and performance evaluation.

📊 2. Overburden Removal (Mm³)



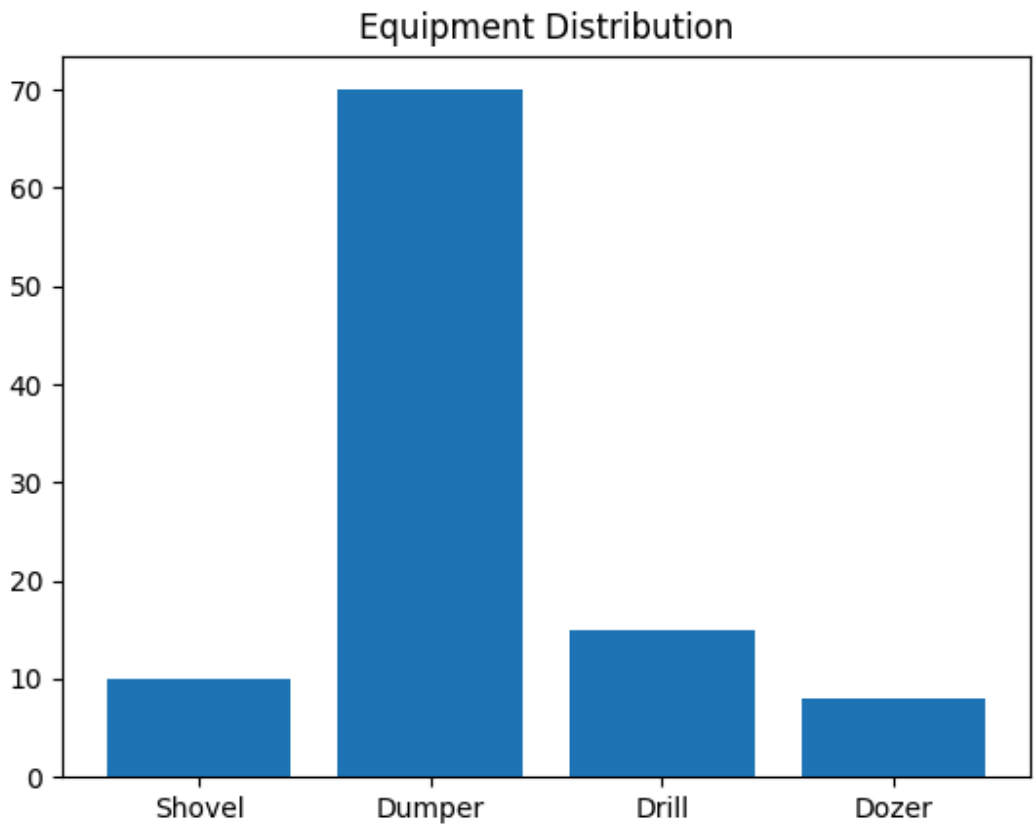
This graph represents the volume of overburden removed to access coal seams.
The increasing trend indicates expansion and deepening of the mine.
Significance: Important for determining stripping ratio, dump design, and equipment requirement.

📊 3. Stripping Ratio Trend



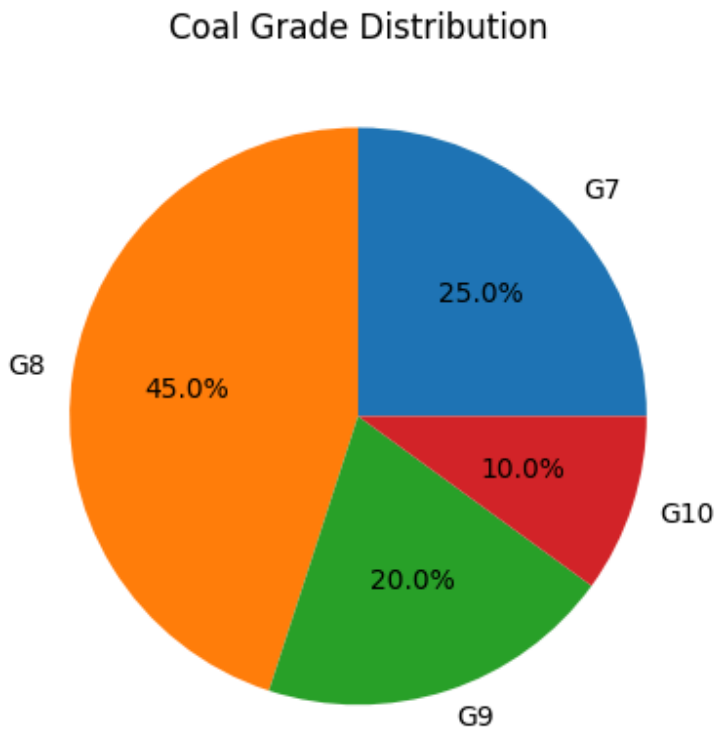
Shows how the stripping ratio changes over time in the mine.
A gradual increase indicates rising mining depth and cost.
Significance: Key economic parameter for assessing mine feasibility and operational efficiency.

📊 4. Equipment Distribution



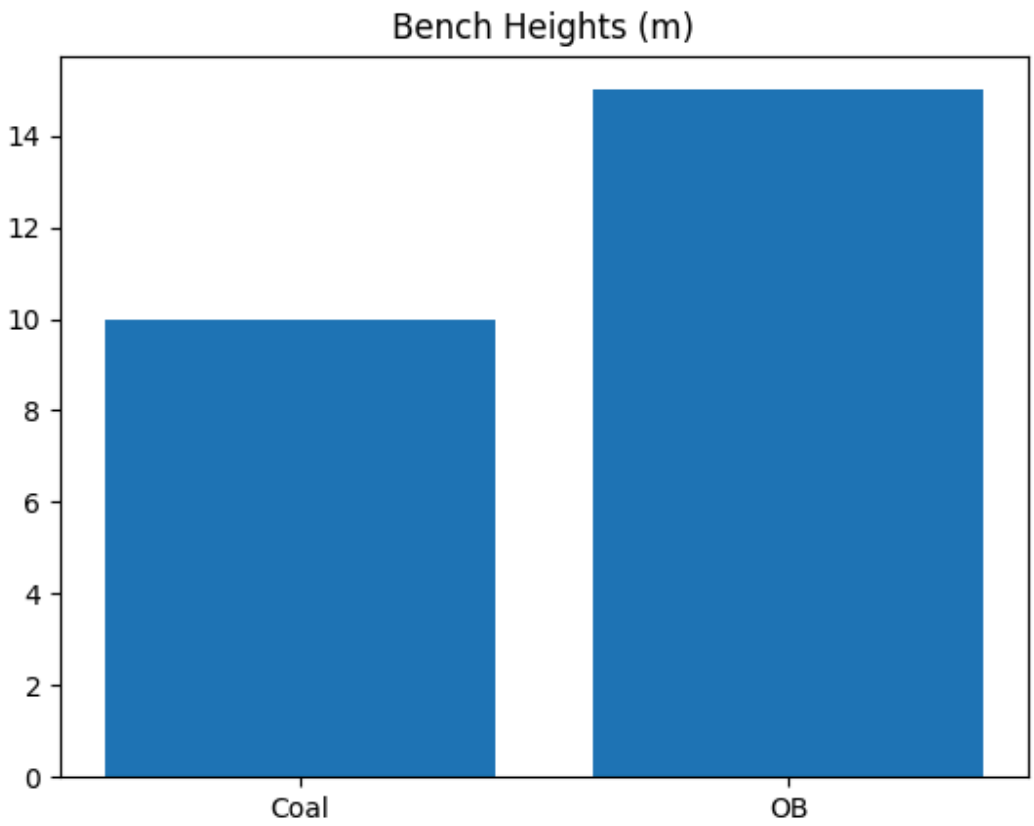
Displays the distribution of major equipment like shovels, dumpers, drills, and dozers.
Dumpers dominate due to large-scale material transportation needs.
Significance: Helps in equipment planning, fleet optimization, and productivity improvement.

📊 5. Coal Grade Distribution



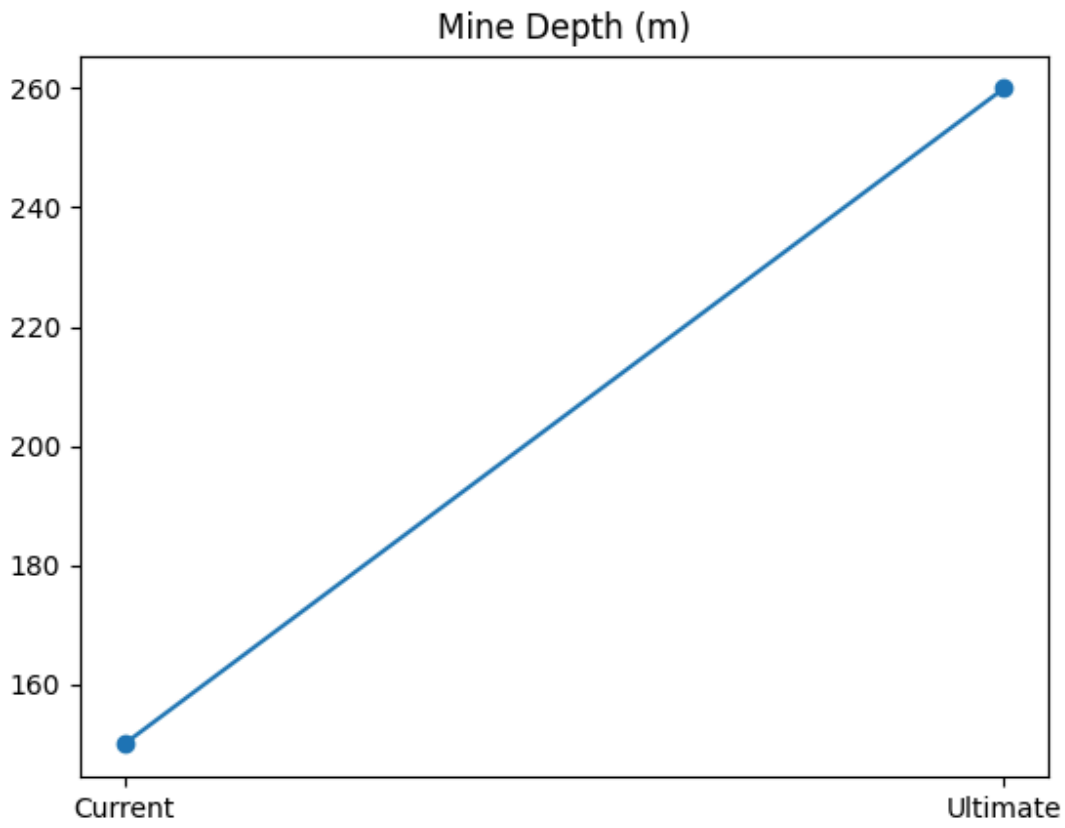
This chart shows the proportion of different coal grades (G7–G10) produced.
Indicates the quality and calorific value of coal from the mine.
Significance: Important for pricing, blending, and supply to thermal power plants.

📊 6. Bench Heights (m)



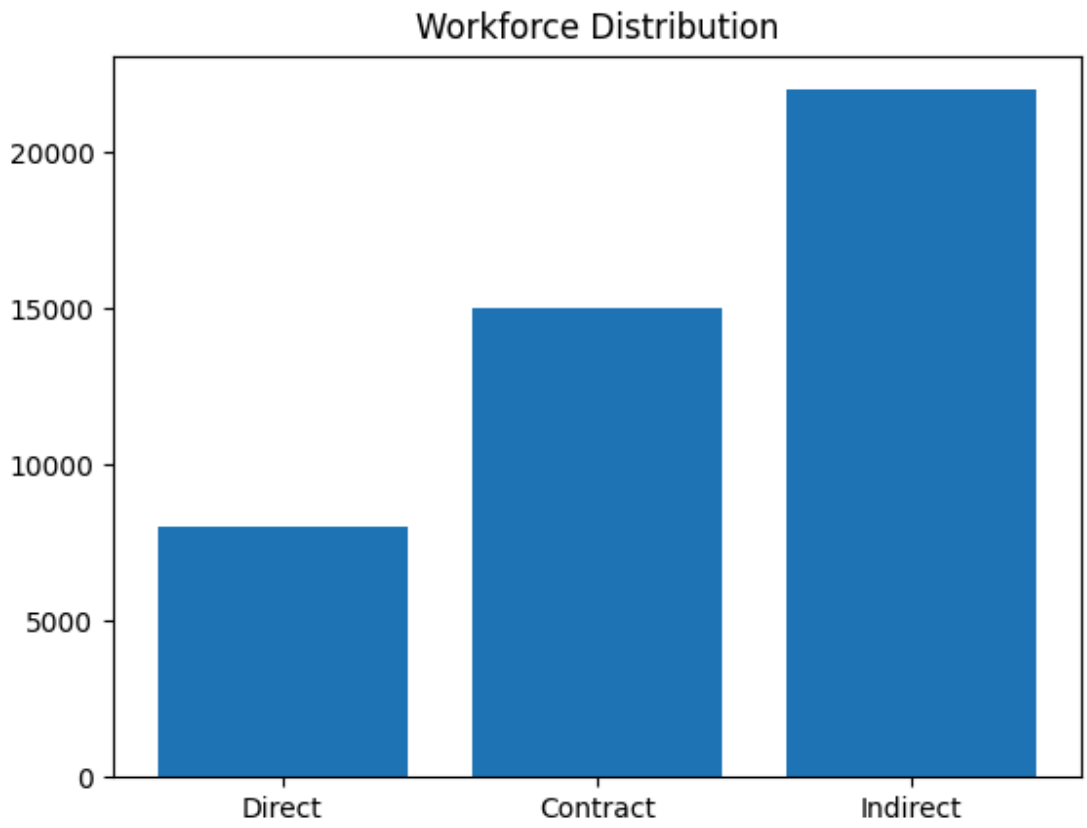
Represents the height of benches used in coal and overburden extraction.
Proper bench design ensures efficient excavation and safety.
Significance: Crucial for slope stability, equipment compatibility, and safe mining operations.

📊 7. Mine Depth Progression



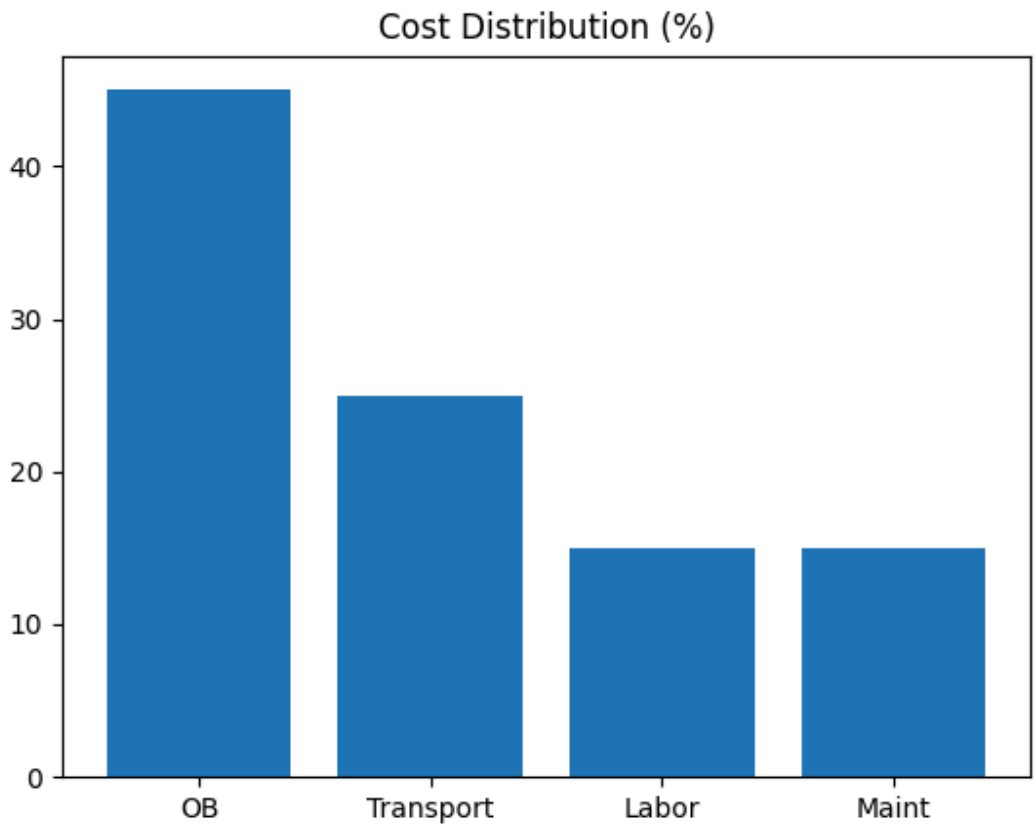
Shows current and ultimate depth of the mine.
Indicates future expansion and deepening of operations.
Significance: Important for long-term mine planning, slope design, and groundwater control.

📊 8. Workforce Distribution



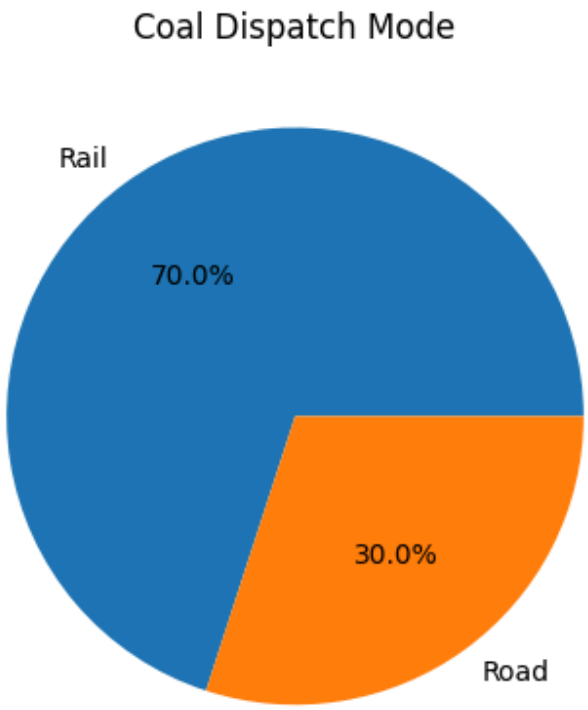
Displays manpower distribution into direct, contract, and indirect categories.
Highlights employment generated by the mining project.
Significance: Useful for manpower planning and socio-economic impact analysis.

📊 9. Cost Distribution (%)



Shows the breakup of mining costs into overburden removal, transport, labor, and maintenance. OB removal generally contributes the highest cost.
Significance: Helps in cost optimization, budgeting, and improving operational efficiency.

📊 10. Coal Dispatch (Rail vs Road)



Represents the proportion of coal transported via rail and road.
Rail transport dominates due to bulk handling efficiency.
Significance: Important for logistics planning, cost reduction, and minimizing environmental impact.